RHEL INTENSIVE NETWORK ADNINISTRATION NOTES:

Lec 23.1

***lscpci**:* to list all pci devices connect with system

```
[root@localhost Downloads]# # lspci <to list all pci devices>
[root@localhost Downloads]# lspci
00:00.0 Host bridge: Intel Corporation Xeon E3-1200 v5/E3-1500 v5/6th Gen Core Pro
cessor Host Bridge/DRAM Registers (rev 07)
00:01.0 PCI bridge: Intel Corporation 6th-10th Gen Core Processor PCIe Controller
(x16) (rev 07)
00:01.2 PCI bridge: Intel Corporation Xeon E3-1200 v5/E3-1500 v5/6th Gen Core Proc
essor PCIe Controller (x4) (rev 07)
00:02.0 VGA compatible controller: Intel Corporation HD Graphics 530 (rev 06)
00:14.0 USB controller: Intel Corporation 100 Series/C230 Series Chipset Family US
B 3.0 xHCI Controller (rev 31)
00:14.2 Signal processing controller: Intel Corporation 100 Series/C230 Series Chi
pset Family Thermal Subsystem (rev 31)
00:16.0 Communication controller: Intel Corporation 100 Series/C230 Series Chipset
 Family MEI Controller #1 (rev 31)
00:16.3 Serial controller: Intel Corporation 100 Series/C230 Series Chipset Family
 KT Redirection (rev 31)
00:17.0 SATA controller: Intel Corporation HM170/QM170 Chipset SATA Controller [AH
CI Mode] (rev 31)
00:1c.0 PCI bridge: Intel Corporation 100 Series/C230 Series Chipset Family PCI Ex
press Root Port #1 (rev f1)
00:1f.0 ISA bridge: Intel Corporation QM170 Chipset LPC/eSPI Controller (rev 31)
00:1f.2 Memory controller: Intel Corporation 100 Series/C230 Series Chipset Family
 Power Management Controller (rev 31)
00:1f.3 Audio device: Intel Corporation 100 Series/C230 Series Chipset Family HD A
```

***lscpci -vvv grep -i ethernet**:* will select only ethernet controller

```
[root@localhost Downloads]# lspci -vvv | grep -i ether
00:1f.6 Ethernet controller: Intel Corporation Ethernet Connection (2) I219-LM (rev 31)
[root@localhost Downloads]# lspci -vvv | grep -i audio
00:1f.3 Audio device: Intel Corporation 100 Series/C230 Series Chipset Family HD Audio Controller (r
ev 31)
[root@localhost Downloads]#
```

ifconfig -a: will show ethernet connections along with their adapter names, in older versions they are named as eth0 eth1 eth2. In Rhel9 naming scheme has changed.

```
[root@localhost Downloads]# #each ethernet adaptor is assigned an Driver
[root@localhost Downloads]# # Drivers eth0 eth1 eth2 eth3 eth4
[root@localhost Downloads]# ifconfig -a
enp0s31f6: flags=4099<UP,BROADCAST,MULTICAST>  mtu 1500
ethernet        ether c8:5b:76:67:13:23  txqueuelen 1000  (Ethernet)
        RX packets 0  bytes 0 (0.0 B)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 0  bytes 0 (0.0 B)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0
        device interrupt 16  memory 0xf2100000-f2120000

lo: flags=73<UP,LOOPBACK,RUNNING>  mtu 65536
        inet 127.0.0.1  netmask 255.0.0.0
Loop back       inet6 ::1  prefixlen 128  scopeid 0x10<host>
        loop  txqueuelen 1000  (Local Loopback)
        RX packets 875  bytes 74406 (72.6 KiB)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 875  bytes 74406 (72.6 KiB)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0
Wifi
wlp3s0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST>  mtu 1500
        inet 192.168.1.36  netmask 255.255.255.0  broadcast 192.168.1.255
        inet6 fe80::6415:b5e2:edfa:628d  prefixlen 64  scopeid 0x20<link>
        ether e4:a7:a0:8e:e4:5f  txqueuelen 1000  (Ethernet)
```

LinkedIn.com/Shujah Ullah
Shujah Ullah
Digital Control Systems || OT || IT || AI || ML ||

Here's what these adapter names represent:

1. **wlp3s0** - This is the name for a wireless network adapter. The "wlp" prefix stands for "Wireless LAN Port".
2. **enp0s31f6** - This is the name for a wired Ethernet network adapter. The "enp" prefix stands for "Ethernet Network Port". The numbers and letters after the prefix provide more information about the physical location of the network card on the motherboard.
3. **lo** - This is the "loopback" interface, which is a virtual network interface used for local communication within the same machine. It's always present and represents the localhost.

The older naming convention used `eth0`, `eth1`, `eth2`, etc. for Ethernet adapters, but the modern naming scheme provides more information about the physical location of the network interfaces, which can be helpful for system administrators and troubleshooting.

More about Naming schemes for network interfaces:

Network interface naming schemes or policies refer to the conventions used by operating systems to automatically assign names to network interfaces on a system. The naming conventions have evolved over time to provide more consistent and informative names.

The most common network interface naming schemes are:

1. **Traditional naming scheme**:
   - This was the original naming scheme used in older Linux distributions.
   - Network interfaces were typically named `eth0`, `eth1`, `eth2`, etc., based on the order in which they were detected by the system.
   - This scheme could be problematic when interfaces were added or removed, as the names might change unexpectedly.
2. **Predictable network interface names (PNIS)**:
   - This naming scheme was introduced in systemd, a system and service manager for Linux.
   - The goal was to provide more predictable and persistent network interface names, even when hardware or configuration changes occur.
   - The names are based on various attributes of the network interface, such as the physical location, firmware/BIOS-provided index, or the bus address.
   - Examples of PNIS names include `enp0s31f6`, `wlp2s0`, `ens3`, etc.

.

LinkedIn.com/Shujah Ullah
Shujah Ullah
Digital Control Systems || OT || IT || AI || ML ||

3. **Consistent Network Device Naming (CNDN)**:
   - This is an alternative naming scheme developed by Dell, and it is used in some Linux distributions, such as Red Hat Enterprise Linux (RHEL) and CentOS.
   - CNDN names follow a similar pattern to PNIS, but the names are based on the physical location of the network interface on the motherboard.
   - Examples of CNDN names include `eno1`, `ens1`, `enx78e7d1234567`, etc.

4. **Netplan-based naming**:
   - This is a newer naming scheme introduced with the Netplan network configuration tool in Ubuntu 17.10 and later.
   - Netplan-based names follow a similar pattern to PNIS and CNDN, but they can also include user-friendly names, such as `ethernet` or `wifi`.
   - Examples of Netplan-based names include `enp0s31f6`, `wlan0`, `ethernet-enp0s31f6`, etc.

The choice of network interface naming scheme can depend on the Linux distribution, the system hardware, and the preferences of the system administrator. The predictable and consistent naming schemes (PNIS, CNDN, and Netplan-based) are generally preferred over the traditional `ethX` naming, as they provide more stability and easier identification of network interfaces.

Ethtool <adaptername>: to list settings for adapter

```
[root@localhost Downloads]# ethtool enp0s31f6
Settings for enp0s31f6:
        Supported ports: [ TP ]
        Supported link modes:   10baseT/Half 10baseT/Full
                                100baseT/Half 100baseT/Full
                                1000baseT/Full
        Supported pause frame use: Symmetric Receive-only
        Supports auto-negotiation: Yes
        Supported FEC modes: Not reported
        Advertised link modes:  10baseT/Half 10baseT/Full
                                100baseT/Half 100baseT/Full
                                1000baseT/Full
        Advertised pause frame use: Symmetric Receive-only
        Advertised auto-negotiation: Yes
        Advertised FEC modes: Not reported
        Speed: Unknown!
        Duplex: Unknown! (255)
        Auto-negotiation: on
        Port: Twisted Pair
        PHYAD: 2
        Transceiver: internal
        MDI-X: Unknown (auto)
        Supports Wake-on: pumbg
        Wake-on: g
        Current message level: 0x00000007 (7)
                               drv probe link
        Link detected: no
```

.

LinkedIn.com/Shujah Ullah
Shujah Ullah
Digital Control Systems || OT || IT || AI || ML ||

Change IP address and netmask by using ifconfig command:

```
[root@localhost Downloads]# ifconfig enp0s31f6 192.168.1.103 netmask 255.255.255.0
[root@localhost Downloads]# #ifconfig
[root@localhost Downloads]# ifconfig
enp0s31f6: flags=4099<UP,BROADCAST,MULTICAST>  mtu 1500
        inet 192.168.1.103  netmask 255.255.255.0  broadcast 192.168.1.255
        ether c8:5b:76:67:13:23  txqueuelen 1000  (Ethernet)
        RX packets 0  bytes 0 (0.0 B)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 0  bytes 0 (0.0 B)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0
        device interrupt 16  memory 0xf2100000-f2120000
```

In the previous RHEL versions IP settings were saved in ifcfg file <etc/sysconfig/network-scripts/ifcfg> However in recent versions ifcfg format is ***deprecated***

```
[root@localhost Downloads]# cd /etc/sysconfig/network-scripts/
[root@localhost network-scripts]# ls
readme-ifcfg-rh.txt
[root@localhost network-scripts]# ^C
[root@localhost network-scripts]# cat readme-ifcfg-rh.txt
NetworkManager stores new network profiles in keyfile format in the
/etc/NetworkManager/system-connections/ directory.

Previously, NetworkManager stored network profiles in ifcfg format
in this directory (/etc/sysconfig/network-scripts/). However, the ifcfg
format is deprecated. By default, NetworkManager no longer creates
new profiles in this format.
```

Im using Rhel 9 and here network configurations are saved in **/etc/NetworkManager/system-connections.**

```
[root@Eagle shujah]# ls /etc/NetworkManager/system-connections
 ethernet1.nmconnection  'The Falcon.nmconnection'
[root@Eagle shujah]#
```

Making Persistent IP changes: above ip remains changed until system doesn't restart. To make persistent IP we have to modify setting in network configuration file.Change autoconnect = false to auto connect = true and save.

```
root@Eagle shujah]# cat /etc/NetworkManager/system-connections/ethernet1.nmconnection
connection]
d=ethernet1
uid=8010aa57-fb78-42f6-8f50-aa13cc83fcc3
ype=ethernet
utoconnect=false
nterface-name=enp0s31f6
imestamp=1716595382

ethernet]
ac-address=C8:5B:76:67:13:23

ipv4]
ddress1=192.168.1.120/24,192.168.1.1
ethod=manual

ipv6]
addr-gen-mode=default
od=ignore

y]
@Eagle shujah]#
```

LinkedIn.com/Shujah Ullah
Shujah Ullah
Digital Control Systems || OT || IT || AI || ML ||

Make connection Active:
*After that type **ifconfig** command in terminal, if your connection isnt showing it means its down check all connections by **ifconfig -a***
*make connection up: **ifconfing enp0s31f6 up***
Check again ifconfig your connection should show if its active.

Restart Network Manager:
to update changes made in config file restart service.
systemctl restart NetworkManager

Changing Network Settings using Network Manager nmcli / nmtui

Using DHCP : Automatic IP Assignment.

nmcli coonection modify <connectionname> ipv4.method auto

```
[root@localhost /]# #Using nmcli for NW config
[root@localhost /]# nmcli connection show
NAME        UUID                                  TYPE      DEVICE
The Falcon  e722fd4a-3418-4bf6-9ddc-0f02d80a5542  wifi      wlp3s0
lo          43625f70-e496-4087-a4e5-d2a51a4fd97c  loopback  lo
ethernet1   8010aa57-fb78-42f6-8f50-aa13cc83fcc3  ethernet  --
[root@localhost /]# # to use DHCP
[root@localhost /]# nmcli connection modify ethernet1 ipv4.method auto
```

Using Static : Manual IP Assignment.

nmcli coonection modify <connectionname> ipv4.method manual ipv4.addresses <ipaddrss/netmask> ipv4.gateway <gate way ip address>

```
[root@Eagle shujah]# ls /etc/NetworkManager/system-connections
 ethernet1.nmconnection  'The Falcon.nmconnection'
[root@Eagle shujah]#
```

Making Persistent IP changes: above ip remains changed until system doesn't restart. To make persistent IP we have to modify setting in network configuration file. Change autoconnect = false to auto connect = true and save.

```
root@Eagle shujah]# cat /etc/NetworkManager/system-connections/ethernet1.nmconnection
connection]
d=ethernet1
uid=8010aa57-fb78-42f6-8f50-aa13cc83fcc3
ype=ethernet
utoconnect=false
nterface-name=enp0s31f6
imestamp=1716595382

ethernet]
ac-address=C8:5B:76:67:13:23

ipv4]
ddress1=192.168.1.120/24,192.168.1.1
ethod=manual

ipv6]
addr-gen-mode=default
ethod=ignore

y]
t@Eagle shujah]#
```

LinkedIn.com/Shujah Ullah
Shujah Ullah
Digital Control Systems || OT || IT || AI || ML ||

Activate your connection:

nmcli connection up <connection name>

```
[root@localhost /]# # activate the profile
[root@localhost /]# nmcli connection show
NAME        UUID                                  TYPE      DEVICE
The Falcon  e722fd4a-3418-4bf6-9ddc-0f02d80a5542  wifi      wlp3s0
lo          43625f70-e496-4087-a4e5-d2a51a4fd97c  loopback  lo
ethernet1   8010aa57-fb78-42f6-8f50-aa13cc83fcc3  ethernet  --
[root@localhost /]# nmcli connection up ethernet1
Connection successfully activated (D-Bus active path: /org/freedesktop/NetworkManager/ActiveConnecti
on/10)
[root@localhost /]# ifconfig
enp0s31f6: flags=4099<UP,BROADCAST,MULTICAST>  mtu 1500
        inet 192.168.1.120  netmask 255.255.255.0  broadcast 192.168.1.255
        ether c8:5b:76:67:13:23  txqueuelen 1000  (Ethernet)
        RX packets 0  bytes 0 (0.0 B)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 0  bytes 0 (0.0 B)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0
        device interrupt 16  memory 0xf2100000-f2120000
```

Restart Network Manager:

to update changes made in config file restart service.

systemctl restart NetworkManager

ip address show <connection name>

```
[root@localhost /]# #verify ip settings of NIC in my case ethernet port isnt connected with router
[root@localhost /]# # so it will show down state
[root@localhost /]# ip address show enp0s31f6
2: enp0s31f6: <NO-CARRIER,BROADCAST,MULTICAST,UP> mtu 1500 qdisc fq_codel state DOWN group default q
len 1000
    link/ether c8:5b:76:67:13:23 brd ff:ff:ff:ff:ff:ff
    inet 192.168.1.120/24 brd 192.168.1.255 scope global noprefixroute enp0s31f6
       valid_lft forever preferred_lft forever
```

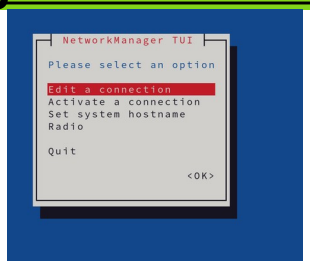
Show Gateway IP:

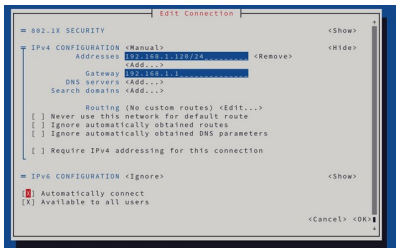
ip route show default

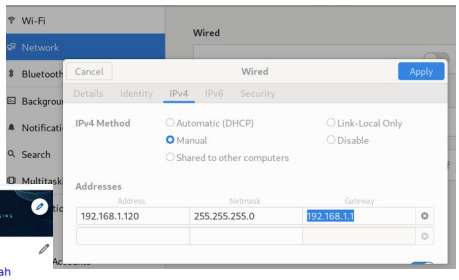
```
[root@localhost /]# ip route show default
default via 192.168.1.1 dev enp0s31f6 proto static metric 100 linkdown
default via 192.168.1.1 dev wlp3s0 proto dhcp src 192.168.1.36 metric 600
[root@localhost /]#
```

using Network Manger GUI interface nmtui

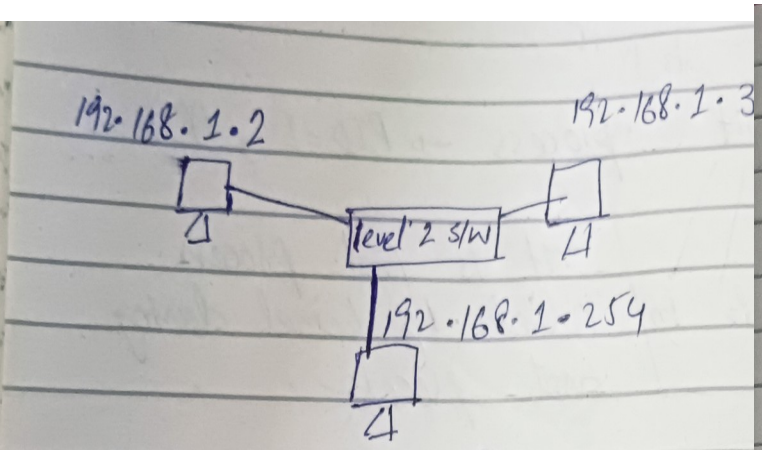
type ***nmtui*** command in terminal and GUI window will open you can also change you network settings from there:







LinkedIn.com/Shujah Ullah
Shujah Ullah
Digital Control Systems || OT || IT || AI || ML ||



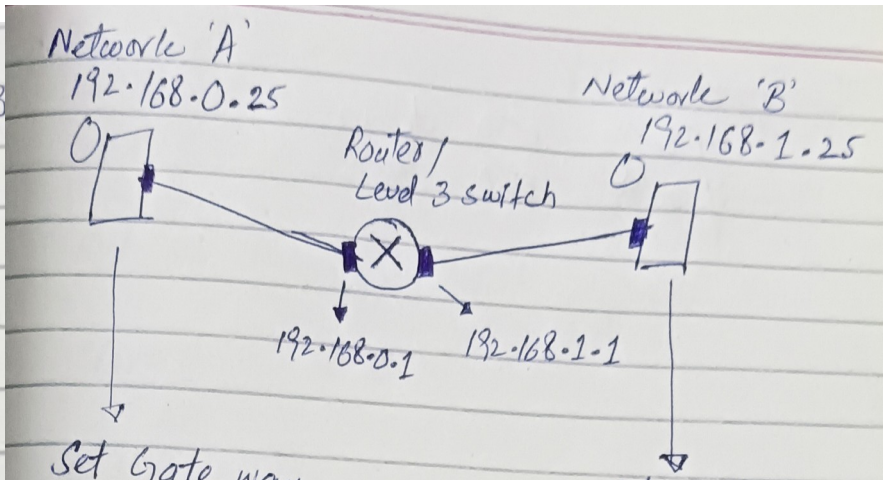
192 . 168 . 1 . 1

Network ~~Address~~

Host IP ~~Address~~

192.168.1.1 → Network Adress-

192.168.1.255 → ~~loop back Add~~ Broad cast IP



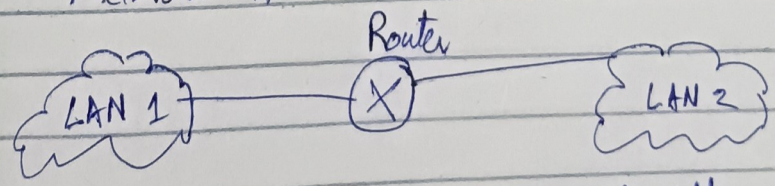
Set Gate way 192.168.0.1

Set Gate way 192.168.1.1

#. Note Router oneside IP is used as Gateway Address to host Network -

Note Router ~~Side~~ Port which is Connected to Network is assigned Network Address of that Network as IP -

"Routers are used to connect two different physical Networks - helps to jump from one Network to other."



route -n #shows GW IP Addrs

route add default gw 192.168.1.1

LinkedIn.com/Shujah Ullah
Shujah Ullah
Digital Control Systems || OT || IT || AI || ML ||